

LABSHEET - 9

HIMANI CHAUDHARY

BTECH(CSE) 3RD YEAR

CU240250967

SECTION - B

ROLL NO. - 24

COMPUTER VISION

```
!pip install ultralytics
```

Collecting ultralytics

Downloading ultralytics-8.4.154-py3-none-any.whl.metadata (46 kB)

46.5/46.5 kB 2.3 MB/s eta 0:00:00

Requirement already satisfied: cloudpickle>=3.1.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (3.1.2)
 Requirement already satisfied: filelock>=3.16.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (3.32.5)
 Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (2.1.3)
 Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (3.10.0)
 Requirement already satisfied: opencv-python!=4.13.0.90,>=4.7.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (4.11.0)
 Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (11.3.0)
 Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (6.0.3)
 Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (2.32.4)
 Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (2.11.0+cpu)
 Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (0.26.0+cpu)
 Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (5.9.5)
 Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (1.35.2)

Collecting nvidia-ml-py>=12.0.0 (from ultralytics)

Downloading nvidia_ml_py-13.610.43-py3-none-any.whl.metadata (9.7 kB)

Collecting ultralytics-thop>=2.1.6 (from ultralytics)

Downloading ultralytics_thop-2.1.6-py3-none-any.whl.metadata (13 kB)

Collecting ultralytics-platform>=0.1.45 (from ultralytics)

Downloading ultralytics_platform-0.1.46-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.3.0)
 Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (4.55.0)
 Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.4.6)
 Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (24.1)
 Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.1.4)
 Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0.post0)
 Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.13/dist-packages (from polars>=0.20.0->ultralytics) (1.35.2)
 Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultralytics) (3.3.0)
 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultralytics) (3.10)
 Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultralytics) (2.2.3)
 Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultralytics) (2025.1.1)
 Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (4.12.0)
 Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (80.9.0)
 Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (1.13.3)
 Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (3.4.1)
 Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (3.1.6)
 Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (2025.1.0)
 Requirement already satisfied: httpx<1,>=0.28 in /usr/local/lib/python3.13/dist-packages (from ultralytics-platform>=0.1.45->ultralytics) (0.28.1)
 Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.28->ultralytics-platform>=0.1.45->ultralytics) (4.6.2.post1)
 Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.28->ultralytics-platform>=0.1.45->ultralytics) (1.0.7)
 Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1,>=0.28->ultralytics-platform>=0.1.45->ultralytics) (0.14.0)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)
 Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3.0)
 Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from jinja2->torch>=1.8.0->ultralytics) (3.0.2)

Downloading ultralytics-8.4.154-py3-none-any.whl (1.4 MB)

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Downloading nvidia_ml_py-13.610.43-py3-none-any.whl (53 kB)

53.2/53.2 kB 4.6 MB/s eta 0:00:00

Downloading ultralytics_platform-0.1.46-py3-none-any.whl (78 kB)

78.1/78.1 kB 7.0 MB/s eta 0:00:00

Downloading ultralytics_thop-2.1.6-py3-none-any.whl (30 kB)

Installing collected packages: nvidia-ml-py, ultralytics-thop, ultralytics-platform, ultralytics

Successfully installed nvidia-ml-py-13.610.43 ultralytics-8.4.154 ultralytics-platform-0.1.46 ultralytics-thop-2.1.6

```
# EXPERIMENT NO. 9
# OBJECT DETECTION AND RECOGNITION USING DEEP LEARNING
# MODEL: YOLOv8
# =====
```

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from ultralytics import YOLO
```

Creating new Ultralytics Settings v0.0.8 file 

View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'

Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com>

1. LOAD PRE-TRAINED YOLOv8 MODEL

=====

```
print("\n=====")
print("OBJECT DETECTION USING YOLOv8")
print("=====")
```

```
print("\nLoading YOLOv8 model...")
```

```
model = YOLO("yolov8n.pt")
```

```
print("YOLOv8 model loaded successfully.")
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
OBJECT DETECTION USING YOLOv8
=====
```

```
Loading YOLOv8 model...
```

```
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt to 'yolov8n.pt': 100% ————— 6.2
```

```
YOLOv8 model loaded successfully.
```

```
Himani Chaudhary
CU240250967
```

2. LOAD IMAGE

=====

IMPORTANT:

Use r before the path to avoid Windows backslash errors.

```
image_path = "/content/beautiful-nature-mountain-scenery-with-flowers-picjumbo-com.jpg"
image = cv2.imread(image_path)
```

```
if image is None:
    print("\nERROR: Image could not be loaded.")
    print("Please check the image path.")
    exit()
```

```
print("\nImage loaded successfully.")
print("Image Size:", image.shape)
print("Himani Chaudhary")
print("CU240250967")
```

```
Image loaded successfully.
Image Size: (3584, 5376, 3)
Himani Chaudhary
CU240250967
```

3. CONVERT IMAGE FROM BGR TO RGB

=====

```
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
print("Himani Chaudhary")
print("CU240250967")
```

```
Himani Chaudhary
CU240250967
```

4. IMAGE PREPROCESSING

=====

```
processed_image = cv2.resize(image_rgb, (640, 640))
```

```
print("Image preprocessing completed.")
print("Himani Chaudhary")
print("CU240250967")
```

```
Image preprocessing completed.
Himani Chaudhary
CU240250967
```

```
# 5. PERFORM OBJECT DETECTION
# =====

print("\nDetecting objects...")

# YOLO automatically performs the required resizing/preprocessing.
results = model(image_rgb)

print("Object detection completed.")
print("Himani Chaudhary")
print("CU240250967")
```

Detecting objects...

0: 448x640 (no detections), 510.6ms
 Speed: 16.5ms preprocess, 510.6ms inference, 11.9ms postprocess per image at shape (1, 3, 448, 640)
 Object detection completed.
 Himani Chaudhary
 CU240250967

```
# 6. GET DETECTION RESULT
# =====

result = results[0]

# Create image with bounding boxes
detected_image = result.plot()

# Convert BGR to RGB for Matplotlib
detected_image_rgb = cv2.cvtColor(
    detected_image,
    cv2.COLOR_BGR2RGB
)
print("Himani Chaudhary")
print("CU240250967")
```

Himani Chaudhary
 CU240250967

```
# 7. PRINT DETECTION RESULTS
# =====

print("\n=====")
print("DETECTED OBJECTS")
print("=====")

object_count = 0

for box in result.boxes:

    object_count += 1

    # Class ID
    class_id = int(box.cls[0])

    # Object name
    class_name = model.names[class_id]

    # Confidence score
    confidence = float(box.conf[0])

    # Bounding box coordinates
    x1, y1, x2, y2 = box.xyxy[0].tolist()

    print("\nObject", object_count)
    print("Class      :", class_name)
    print("Confidence  :", round(confidence, 2))

    print(
        "Bounding Box:",
        "(",
        round(x1, 2),
        ", ",
        round(y1, 2),
        ", ",
        round(x2, 2),
        ", ",
        round(y2, 2),
        ")",
```

```

        round(y2, 2),
        ")")
    )
    print("Himani Chaudhary")
    print("CU240250967")

```

```

=====
DETECTED OBJECTS
=====

```

```

# 8. TOTAL OBJECTS
# =====

print("\n=====")
print("TOTAL OBJECTS DETECTED:", object_count)
print("=====")
print("Himani Chaudhary")
print("CU240250967")

```

```

=====
TOTAL OBJECTS DETECTED: 0
=====
Himani Chaudhary
CU240250967

```

```

# 9. DISPLAY ORIGINAL IMAGE
# =====

plt.figure(figsize=(10, 7))

plt.imshow(image_rgb)

plt.title("Original Image")

plt.axis("off")

plt.show()
print("Himani Chaudhary")
print("CU240250967")

```

Original Image



Himani Chaudhary
CU240250967

```

# 10. DISPLAY DETECTION RESULT
# =====

plt.figure(figsize=(10, 7))

```

```
plt.imshow(detected_image_rgb)

plt.title("YOLOv8 Object Detection Result")

plt.axis("off")

plt.show()
print("Himani Chaudhary")
print("CU240250967")
```

YOLOv8 Object Detection Result



Himani Chaudhary
CU240250967

```
# 11. SAVE DETECTION RESULT
# =====

output_path = "/content/beautiful-nature-mountain-scenery-with-flowers-picjumbo-com.jpg"

cv2.imwrite(
    output_path,
    detected_image
)

print("\nDetection result saved successfully.")
print("Saved at:", output_path)
print("Himani Chaudhary")
print("CU240250967")
```

Detection result saved successfully.
Saved at: /content/beautiful-nature-mountain-scenery-with-flowers-picjumbo-com.jpg
Himani Chaudhary
CU240250967

```
# 12. EVALUATION
# =====

print("\n=====")
print("EVALUATION")
print("=====")

print("Number of detected objects:", object_count)

print("\nThe detected objects are displayed with:")

print("1. Bounding boxes")
print("2. Class labels")
print("3. Confidence scores")
print("\nExperiment completed successfully.")
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
EVALUATION
=====
Number of detected objects: 0

The detected objects are displayed with:
1. Bounding boxes
2. Class labels
3. Confidence scores

Experiment completed successfully.
Himani Chaudhary
CU240250967
```